

Merit1st (SC OS3) — Complete User & Administrator Guide

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1. App Overview

Merit1st (internal name: SC OS3) is a floating overlay camera application for Android. It renders a small, draggable camera preview on top of any other app, allowing you to capture photos discreetly without switching away from your current application.

Property	Value
App name	Merit1st
Package	com.scos3.camera
Version	0.1.0 (versionCode 1)
Min Android	10 (API 29)
Target Android	16 (API 36)
Camera	CameraX + Camera2
Face detection	Google ML Kit
Email	Gmail SMTP (JavaMail)

Key features:

- Floating camera preview overlay (draggable, resizable)

- Single photo capture with autofocus

- Burst capture (5 consecutive photos)

- Auto / interval capture (2s, 4s, 6s, 10s intervals)

- Face-detection auto capture

- Front/back camera switch

- Ultra-wide (0.5x) camera support where available

- Zoom slider

- Email delivery via Gmail SMTP

- Save to Gallery or private storage

- Volume button shortcuts (via accessibility service)

- Background capture with notification

2. Installation & Permissions

Required Permissions

When you first open the app, it will request these permissions:

Permission	Why	How to Grant
Camera	Takes photos	Tap "Allow camera" in the Android Setting: Permissions
Notifications	Shows capture status in notification bar	Allow when prompted
Display over other apps	Shows the floating overlay on top of other apps	Tap "Open settings" on the permission card to enable

If Camera Permission Does Not Work

If tapping "Allow camera" does not respond:

- Go to Android **Settings** > **Apps** > **Merit1st** > **Permissions**
- Tap **Camera** and select **Allow**
- Force-close and reopen the app

Via ADB (for developers/testers)

```
“ bash
```

```
adb shell pm grant com.scos3.camera android.permission.CAMERA
```

```
adb shell pm grant com.scos3.camera android.permission.POST_NOTIFICATIONS
```

```
adb shell am start -n com.scos3.camera/.ui.camera.MainActivity
```

3. Main Interface — Overlay System

Merit1st uses two floating overlay windows that appear on top of your current app:

Upper Overlay (top of screen)

+-----+	
[BLACK] [SETTING]	
[SWITCH] [HELP]	
+-----+	
<- Camera Preview	
Live Preview	(240x320dp)
(draggable)	
+-----+	
"Merit1st - drag"	
<--SIZE slider--> <--ZOOM slider-->	
+-----+	
Ready	<- Status chip
+-----+	
+-----+	
`	

Lower Overlay (bottom of screen)

A diagram showing a 4x4 grid. The columns are labeled BURST, CAPTURE, and AUTO. The rows are labeled row 1, row 2, row 3, and row 4. The grid is divided into four quadrants by a vertical line between CAPTURE and AUTO, and a horizontal line between row 2 and row 3. The top-left quadrant (BURST, row 1) is light blue. The top-right quadrant (CAPTURE, row 1) is light green. The bottom-left quadrant (BURST, row 2) is light orange. The bottom-right quadrant (CAPTURE, row 2) is light purple. The bottom-right quadrant (CAPTURE, row 3) is light yellow. The bottom-right quadrant (CAPTURE, row 4) is light pink.

[MINIMIZE] [EXIT]

+-----+

Dragging the Preview

Touch and hold the **"Merit1st - drag"** label below the camera preview to reposition the entire floating panel anywhere on screen.

Resizing the Preview

Use the **SIZE slider** (left side) to make the camera preview window larger or smaller. Range: 24% to 62% of screen width.

4. All Buttons — Complete Reference

Mode Buttons (Bottom Row)		
Button	Action	Details
CAPTURE	Take one photo	Triggers autofocus captures. Shows Merit1st_yyyyMM Produces a shot
BURST	Take 5 consecutive photos	Captures 5 photos between each. B "STOP" while running as "Burst 1/5", "STOP to cancel.
AUTO	Start/stop interval capture	Starts taking photos interval (2/4/6/1 Runs in foreground even when screen changes to "Stopped" also toggles this
FACE	Enable face-detection capture	When ON, auto whenever a face preview. 3-second captures. Button ON".
Control Buttons (Top Area)		
Button	Action	Details
BLACK	Blackout mode	Hides the camera the camera running discreet capture to show preview "BLACK ON".
SWITCH	Switch camera	Toggles between camera.
SETTING	Open settings	Opens the Settings temporarily hidden
HELP	Show help	Displays a help descriptions. Also diagnostics" button

Bottom Buttons		
Button	Action	Details
MINIMIZE	Minimize to background	Hides the overlay capture running notification showing the notification icon
EXIT	Stop and close	Stops all capture foreground service overlay, and closes the app
Sliders		
Slider	Location	Action
SIZE	Left side	Adjusts the camera size. Default position is 1.0x
ZOOM	Right side	Controls digital zoom (wide) to max zoom readout ("1.0x") toggle between 1.0x and 2.0x
Preview Tap		
Gesture	Action	
Tap the preview	Triggers tap-to-focus at the tapped point. Uses auto-focus + auto-exposure.	

5. Capture Modes

Single Capture (CAPTURE)

- Tap **CAPTURE** or press **Volume Down**
- Camera autofocuses at center of preview (up to 2 seconds)
- Photo is captured as JPEG
- Stored to Gallery or private storage (based on setting)
- Email queued if enabled
- Short vibration confirms capture
- Status shows "Saved: [filename]"

Burst Capture (BURST)

- Tap **BURST** to start
- 5 photos captured consecutively, 350ms apart

- Progress: "Burst 1/5", "Burst 2/5", ...
- Other buttons (CAPTURE, AUTO, SWITCH, SIZE, ZOOM) are disabled during burst

- Tap **STOP** (the BURST button changes to STOP) to cancel mid-burst

- Result: "Burst done - N saved" or "Burst cancelled"

Auto / Interval Capture (AUTO)

- Set the interval in Settings (2s, 4s, 6s, 10s, 12s, 14s) or use the inline interval selector (2s/4s/6s/10s)

- Tap **AUTO** or press **Volume Up**

- A foreground notification appears: "Auto capture active"

- Photos are taken at the selected interval

- Camera continues even when screen is off or app is minimized

- Tap **MINIMIZE** to hide the overlay while auto continues

- Tap the notification or reopen the app, then tap **Stop capture** to stop

- Status: "Auto capture active - every Ns - M saved"

Face Detection Capture (FACE)

- Tap **FACE** to enable (button shows "FACE ON")

- ML Kit face detection analyzes the preview

- When a face is detected, a photo is automatically taken

- 3-second cooldown between captures (prevents rapid-fire)

- Tap **FACE** again to disable

- Works with both front and back cameras

6. Camera Features — Zoom, Focus, Lens

Zoom

- Use the **ZOOM slider** on the right side of the upper overlay

- Range depends on device (typically 0.5x to 10x)
- Tap the zoom readout text ("1.0x") to quickly toggle between ultra-wide (0.5x) and normal (1.0x)
- If your device has an ultra-wide camera, dragging below 1.0x activates it

Ultra-Wide (0.5x) Support

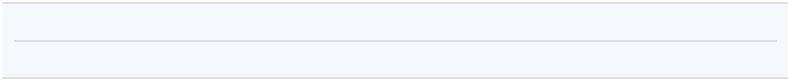
- Automatically detected on devices with a dedicated ultra-wide camera
- Check **Settings** > Ultra-wide info to see if your device supports it
- Also check via **Diagnostics** for detailed camera hardware info

Tap to Focus

- Tap anywhere on the camera preview to focus at that point
- Uses auto-focus + auto-exposure metering

Front/Back Camera

- Tap **SWITCH** to toggle between back and front cameras
- The default camera can be set in **Settings** > Default camera
- Ultra-wide is only available on the back camera



7. Settings — Complete Guide

Access via **SETTING** button on the overlay.

Camera Settings

Setting	Options	Default
Image resolution	AUTO + device-specific sizes	AUTO
Image quality	Low (70), Standard (85), High (95)	High (95)
Default camera	Back camera, Front camera	Back camera

Capture Settings

Setting	Options	Default
---------	---------	---------

Auto capture interval	2s, 4s, 6s, 10s, 12s, 14s	4s
-----------------------	---------------------------	----

Delivery Settings

Setting	Options	Default
Email sending	ON / OFF	OFF
Save photos to Gallery	ON / OFF	ON

Important: If both Email and Gallery are OFF, photos still save to app-private storage and remain available for email if you turn email back on later.

Volume Button Control

Setting	Description
Status	Shows whether the accessibility service is enabled
Open Accessibility Settings	Takes you to Android Accessibility settings to enable "Merit1st volume control"
When enabled:	
Volume Down = Take one photo	
Volume Up = Toggle auto capture	
Volume buttons work normally in all other apps	

Gmail Configuration

Field	Description
Sender email	Your Gmail address (e.g., you@gmail.com)
Sender app password	16-character Gmail App Password (NOT your regular password)
Receiver email	Where to send photos (can be the same as sender)
Test email	Sends a test message to verify the setup
Remove stored credentials	Clears saved Gmail credentials

Other

Button	Description
Open device diagnostics	Shows camera hardware information
Save settings	Saves all settings and closes

8. Email Delivery — Gmail Setup

How It Works

When email is ON, every photo you capture is:

- Compressed to max 2048px, JPEG quality 80 (smaller file for email)
- Added to a persistent queue (survives app restarts)
- Sent via Gmail SMTP (smtp.gmail.com, port 465, SSL)
- If sending fails, it retries with exponential backoff (15s, 30s, 60s, ... up to 5 minutes)
- If network is unavailable, photos stay queued and retry when network returns

Step-by-Step: How to Create a Gmail App Password

IMPORTANT: You CANNOT use your regular Gmail password. You must create a special 16-character "App Password" for Merit1st.

Prerequisites

- A Gmail account (you@gmail.com)
- **2-Step Verification must be enabled** on your Google account

Steps to Enable 2-Step Verification (if not already enabled):

- Open a browser and go to: <https://myaccount.google.com/security>
- Scroll to **"How you sign in to Google"**
- Click **2-Step Verification**
- Follow the prompts to enable it (requires a phone number)
- Verify it is ON

Steps to Create the App Password:

- Go to: <https://myaccount.google.com/apppasswords>
- (If you cannot find it, search "App Passwords" in the Google Account search bar)
- You may be asked to sign in again
- At the bottom, under **"App name"**, type: Merit1st
- Click **"Create"**

- Google will show a **16-character password** like: abcd efgh ijkl mnop

- **Copy this password** (remove the spaces — it becomes abcdefghijklmnop)

- Click **Done**

Save this password. You will only see it once. If you lose it, create a new one at the same page.

How to Enter in Merit1st:

- Open Merit1st > tap **SETTING**

- Under **"Gmail (email delivery)":**

- **Sender email:** Enter your full Gmail address (e.g., you@gmail.com)

- **Sender app password:** Enter the 16-character app password (no spaces)

- **Receiver email:** Enter where you want photos sent (can be the same email)

- Tap **"Test email"** to verify it works

- If "Test email sent" appears, the setup is correct

- Turn **Email sending** ON

- Tap **"Save settings"**

Gmail App Password — Common Errors

Error Message	Cause	Fix
"Gmail authentication failed"	Wrong password or 2-Step Verification not enabled	Re-create the App Password or enable 2-Step Verification
"Network unavailable"	No internet connection	Connect to Wi-Fi or mobile data. Photos stay queued
"Configure sender, app password and receiver first"	Fields are empty	Fill in all three fields

Email Security

- Credentials are stored encrypted on the device using Android Keystore (AES-256-GCM)

- They are never sent to any third-party server — only used with smtp.gmail.com

- Credentials are never written to logs, source code, or build files

- You can clear them anytime via **"Remove stored credentials"**

9. Storage — Where Photos Are Saved

Gallery Mode (Default: ON)

Property	Value
Storage type	Android MediaStore (Gallery-visible)
Folder	Pictures/Merit1st/
Full path	/storage/emulated/0/Pictures/Merit1st/
File naming	Merit1st_yyyyMMdd_HHmssSSS.jpg
Visibility	Visible in Android Gallery, Google Photos, file managers
Example	Merit1st_20260822_143025123.jpg

Private Storage Mode (When Gallery is OFF)

Property	Value
Storage type	App-private files directory
Folder	private_captures/
Full path	/data/data/com.scos3.camera/files/private_captures/
File naming	Merit1st_yyyyMMdd_HHmssSSS.jpg
Visibility	NOT visible in Gallery. Only accessible via ADB or root.
Persistence	Survives app restarts. NOT deleted as cache.

Storage Decision

The app checks the "Save photos to Gallery" setting for every capture:

- **ON** -> Photos go to MediaStore (Gallery)
- **OFF** -> Photos go to app-private storage

Both paths receive the original JPEG byte-for-byte (no recompression).

Email Compression (Separate from Storage)

When email is ON, the original JPEG is also:

- Compressed to max 2048px / quality 80
- Saved to: /data/data/com.scos3.camera/files/email_captures/
- This is a **copy** — the original photo is never modified
- The compressed copy is deleted after successful email delivery

10. Volume Button Control

What It Does

When enabled, you can use your phone's physical volume buttons to control Merit1st without touching the overlay:

Button	Action
Volume Down	Take one photo (same as CAPTURE button)
Volume Up	Toggle auto capture on/off (same as AUTO button)

How to Enable

- Open Merit1st > tap **SETTING**
- Under "**Volume button control**", tap "**Open Accessibility Settings**"
- Find "**Merit1st volume control**" in the accessibility list
- Toggle it ON
- Confirm the dialog (the service only intercepts volume keys when Merit1st overlay is active)

Important Notes

- Volume buttons work **normally** in all other apps
- They only control Merit1st when the floating overlay is visible on screen
- The service does NOT read screen content, text, or any other app data
- It only intercepts volume key press events
- You can disable it anytime in Accessibility Settings

11. Foreground Service & Background Capture

What Happens When You MINIMIZE

When AUTO capture is running and you tap MINIMIZE:

- The overlay hides (no visible UI)
- A foreground notification appears: "Merit1st - Auto capture active"
- The camera stays open via a foreground service
- Photos continue to be taken at the configured interval

- A partial wake lock keeps the CPU running (capture works even when screen is off)

- Tap the notification to return to the overlay

What Happens When You EXIT

Tapping EXIT:

- Stops all capture (single, burst, auto, face)

- Stops the foreground service

- Releases the wake lock

- Removes the overlay

- Closes the app

Notification Controls

The notification shows:

- "Merit1st" title

- "Auto capture active" subtitle

- Current interval and photo count

- **Open** button (returns to overlay)

- **Stop** button (stops capture and service)

12. Device Diagnostics

Access via **SETTING** > "**Open device diagnostics**" or via **HELP** > "**Device diagnostics**".

The diagnostics screen shows:

- Device manufacturer and model

- Android version and SDK level

- All Camera2 camera IDs (facing, logical/physical, capabilities)

- Back camera zoom range

- Ultra-wide camera detection (physical camera ID, FOV delta)

- Sub-unit zoom support (zoom below 1.0x)

- Supported JPEG resolutions (back and front, top 10)

- Overall verdict on ultra-wide support

Use the **Copy** button to copy the report to clipboard, or **Share** to send it.

13. Accessing Private Storage via CMD

When "Save photos to Gallery" is OFF, photos are saved to app-private storage. Here is how to access them.

Via ADB Shell (No Root Required)

```
`bash
```

List all saved photos

```
adb shell ls -la /sdcard/Android/data/com.scos3.camera/files/private_captures/
```

Pull a specific photo to your computer

```
adb pull /sdcard/Android/data/com.scos3.camera/files/private_captures/Merit1st_20260822_143025123.jpg C:\Users\YourName\Pictures\
```

Pull ALL photos at once

```
adb pull /sdcard/Android/data/com.scos3.camera/files/private_captures/ C:\Users\YourName\Merit1st_Backup\
```

Check how many photos are stored

```
adb shell ls /sdcard/Android/data/com.scos3.camera/files/private_captures/
```

```
wc -l
```

Via ADB Shell (Alternative Path for Debug Builds)

```
`bash
```

For debug builds, use run-as to access internal storage

```
adb shell run-as com.scos3.camera ls files/private_captures/
```

Pull from internal storage

```
adb shell run-as com.scos3.camera cat files/private_captures/Merit1st_20260822_143025123.jpg > C:\Users\YourName\Pictures\photo.jpg
```

Via File Manager (Android)

- Open a file manager app (e.g., "Files by Google", "Solid Explorer")
- Navigate to: `Android/data/com.scos3.camera/files/private_captures/`
- Note: On Android 11+ (API 30+), direct access to `Android/data/` **may be restricted**. Use a **file manager with SAF support** or **ADB**.

Viewing Gallery Photos (When Gallery is ON)

Photos are saved to: `/storage/emulated/0/Pictures/Merit1st/`

```
` bash
```

Via ADB

```
adb shell ls /sdcard/Pictures/Merit1st/
```

Pull from Gallery path

```
adb pull /sdcard/Pictures/Merit1st/ C:\Users\YourName\Merit1st_Gallery\
```

Or simply open the Gallery app on your phone — photos appear in the "Merit1st" album

```
`
```

Accessing Email Queue

```
` bash
```

View pending email queue

```
adb shell cat /sdcard/Android/data/com.scos3.camera/files/email_queue/pending.json
```

View compressed email copies

```
adb shell ls /sdcard/Android/data/com.scos3.camera/files/email_captures/
```

Pull compressed email copies

```
adb pull /sdcard/Android/data/com.scos3.camera/files/email_captures/  
C:\Users\YourName\Merit1st_Email\
```

```
`
```

Accessing Shared Preferences (Settings)

```
` bash
```

View app settings (debug builds only)

```
adb shell run-as com.scos3.camera cat shared_prefs/scos3_settings.xml
```

View secure preferences (encrypted, not human-readable)

```
adb shell run-as com.scos3.camera cat shared_prefs/scos3_secure.xml
```

Full Backup via ADB

```
` bash
```

Create a backup of ALL app data

```
adb shell am backup -f /sdcard/scos3_backup.ab com.scos3.camera
```

Pull the backup to computer

```
adb pull /sdcard/scos3_backup.ab C:\Users\YourName\Backup\
```

Accessing ALL App Files via ADB

```
` bash
```

List all files in the app's files directory

```
adb shell ls -R /sdcard/Android/data/com.scos3.camera/
```

Full directory tree

```
adb shell find /sdcard/Android/data/com.scos3.camera/ -type f
```

14. Troubleshooting

Camera Permission Not Working

Symptom: "Allow camera" button does not respond.

Fix:

- Go to Settings > Apps > Merit1st > Permissions
- Enable Camera permission manually

- Force-stop and restart the app

Overlay Not Appearing

Symptom: App opens but no floating overlay is visible.

Fix:

- Go to Settings > Apps > Merit1st > Permissions
- Enable "Display over other apps" / "Appear on top"
- Restart the app

Email Not Sending

Symptom: Photos captured but no email received.

Check:

- Is "Email sending" toggled ON in Settings?
- Are all three Gmail fields filled in (sender, app password, receiver)?
- Is the App Password correct? (16 characters, no spaces)
- Is 2-Step Verification enabled on the Gmail account?
- Is the phone connected to the internet?
- Tap "Test email" to diagnose

Auto Capture Not Working

Symptom: Tap AUTO but nothing happens.

Check:

- Ensure camera permission is granted
- Ensure "Display over other apps" is enabled (needed for foreground service)
- Ensure notification permission is granted (Android 13+)
- Check that another AUTO/burst capture is not already running

Volume Buttons Not Responding

Symptom: Pressing volume keys does not capture.

Check:

- Enable "Merit1st volume control" in Android Accessibility Settings
- Ensure the Merit1st overlay is visible on screen (volume keys only work when overlay is active)
- Volume keys work normally in all other apps

Black Screen / Camera Failed

Symptom: Camera preview shows black.

Fix:

- Check if another app is using the camera (close other camera apps)
- Check camera permission
- Try SWITCH to toggle between front/back camera
- Open Diagnostics to check camera hardware status

App Crashes on Launch

Fix:

- Clear app data: Settings > Apps > Merit1st > Storage > Clear data
- Re-grant permissions
- Reconfigure settings

Photos Not Visible in Gallery

Check:

- Is "Save photos to Gallery" toggled ON in Settings?
- Open the Gallery app and look for the "Merit1st" album
- On some devices, MediaStore may take a few seconds to refresh

15. Technical Reference

App Configuration

Item	Value
Package name	com.scos3.camera
Version name	0.1.0
Version code	1
Min SDK	29 (Android 10)
Target SDK	36 (Android 16)
Compile SDK	36
Java target	JVM 17
Build type	Release (minified + R8)

Permissions (AndroidManifest.xml)

Permission	Purpose
CAMERA	Camera hardware access
INTERNET	Gmail SMTP email delivery
ACCESS_NETWORK_STATE	Network state for email retry
FOREGROUND_SERVICE	Background auto capture
FOREGROUND_SERVICE_CAMERA	Camera while-in-use in foreground service
POST_NOTIFICATIONS	Capture status notifications
VIBRATE	Capture vibration feedback
WAKE_LOCK	Keep CPU alive during auto capture
SYSTEM_ALERT_WINDOW	Floating overlay windows

Default Settings

Setting	Default Value
Resolution	AUTO
Quality	HIGH (95)
Default lens	BACK
Auto interval	4 seconds
Email sending	OFF
Gallery save	ON
Burst count	5 photos
Burst delay	350ms
Haptic vibration	65ms
Face cooldown	3000ms
Focus timeout	2000ms
Email compress max	2048px / quality 80

File Paths Summary

Path	Description
/storage/emulated/0/Pictures/Merit1st/	Gallery photos
files/private_captures/	Private storage photos
files/email_captures/	Compressed email copies
files/email_queue/pending.json	Pending email queue
shared_prefs/scos3_settings.xml	App settings
shared_prefs/scos3_secure.xml	Encrypted Gmail credentials

Email Technical Details

Property	Value
SMTP server	smtp.gmail.com
Port	465 (SSL)
Auth method	Gmail App Password (16-char)
Compress max dimension	2048px
Compress JPEG quality	80
Initial retry delay	15 seconds
Max retry delay	5 minutes

Retry backoff	Exponential (2x)	
Key Constants		
Constant	Value	Location
BURST_COUNT	5	MainActivity.kt
BURST_DELAY_MS	350ms	MainActivity.kt
SHUTTER_TICK_MS	65ms	Haptics.kt
FOCUS_TIMEOUT_MS	2000ms	CameraController.kt
CAPTURE_COOLDOWN_MS	3000ms	FaceDetectionController.kt
FRAME_THROTTLE_MS	250ms	FaceDetectionController.kt
MIN_FACE_RATIO	0.1 (10%)	FaceDetectionController.kt
MAX_DIMENSION	2048px	ImageCompressor.kt
EMAIL_QUALITY	80	ImageCompressor.kt
INITIAL_RETRY_MS	15000ms	EmailManager.kt
MAX_RETRY_MS	300000ms (5 min)	EmailManager.kt

Camera Architecture

Component	Role
CameraManager	ProcessCameraProvider singleton
CameraController	CameraX facade: preview, capture, zoom, lens, focus
CameraCapabilities	Camera2 hardware characteristics, ultra-wide detection
FaceDetectionController	ML Kit face detection state machine
CaptureEngine	Single/burst/interval capture + storage + email
Haptics	Shutter vibration feedback
OverlayManager	Two-part floating overlay WindowManager
SettingsRepository	SharedPreferences persistence
SecurePrefs	Android Keystore encrypted credentials
EmailManager	Queue coordination, retry, network callback
GmailEmailRepository	Gmail SMTP implementation
ImageCompressor	JPEG compression for email
MediaStoreRepository	Gallery-visible storage
PrivateStorageRepository	App-private durable storage
CaptureForegroundService	Background camera service
VolumeKeyAccessibilityService	Volume key interception
VolumeKeyDispatcher	App-level volume key bridge